Key Findings and Data Quality Issues

1.Dataset overview

The train test dataset contains 48,000 labeled load records covering the period from January 1, 2025 to October 31, 2025. There are 14 columns, including route information, geographic coordinates, distance, equipment type, weight, date, market indicators, and the target variable “posted_rate”.

No duplicate complete rows or duplicate “load_id” values were found.

Data quality issues

Several data-quality issues were identified during the initial investigation:

- Missing weight values: 300 rows (0.625% of the dataset) contain missing weight values.
- Invalid weight values: 292 rows contain non-positive weights. Since load weight cannot reasonably be negative or zero, these values were treated as invalid rather than valid observations.
- Missing “market_index” values: 374 rows (0.779%) contain missing market index values.
- No invalid distances: All distance values were positive and within the expected range.
- No invalid target values: all “posted_rate” values are positive.
- No duplicate records: neither complete-row duplicates nor duplicate “load_id” values were found.
- Coordinate consistency: each pickup and delivery city has consistent coordinates across the dataset. No cities with conflicting coordinates were detected.
- Categorical values: equipment contains three equipment types: Dry Van, Reefer, and Flatbed, with no unexpected equipment categories identified.

I held off on immediately dropping the invalid/missing weight rows. Their target distribution looked similar to the rows with valid weights, so it made more sense to keep them and treat the bad weights as missing during preprocessing rather than throw away otherwise-usable data.

Key findings from exploratory analysis

Distance was clearly the strongest driver of rate - Pearson correlation of 0.909 with “posted_rate”. Everything else was much weaker on a linear basis:

- weight: 0.035
- market_index: 0.034
- quote_signal: -0.040

However, weak linear correlation does not necessarily mean that these variables are useless for a machine-learning model, since they may contribute through nonlinear relationships or interactions with other features.

Equipment type also showed a noticeable difference in average rates:

- Dry Van: $2,271.55
- Flatbed: $2,445.09
- Reefer: $2,553.64

So equipment type carries real signal too, just not in a simple linear sense.

Temporal patterns

“posted_rate” moved noticeably over the year — averaging around $2,256 in January, climbing to about $2,497 by June, then dropping back to roughly $2,379 in October.

market_index followed a similar arc, from about 0.93 in January up to 1.30 in May and back down to around 0.96 by October. Meanwhile average route distance barely moved across the same period — so the swings in rate clearly aren't just a function of routes getting longer or shorter. There's a temporal or market-driven component worth capturing.

Validation strategy implication

Because the labeled development data covers January through October and the final validation set represents a later period, a time-based validation strategy is more appropriate than a random split.

The planned approach is to train on earlier months and validate on a later month, for example:

Training: January to September
Validation: October

After selecting and tuning the model, the final model can be trained on all available January–October labeled data and used to generate predictions for the unseen validation dataset.

2. Baseline Model Development & Validation

Feature Engineering

Several additional features were created from the original data:

- month - month extracted from the load date.
- day_of_week - day of the week extracted from the load date.
- route - combination of pickup and delivery cities.

The original categorical features pickup, delivery, and equipment were provided to CatBoost as categorical variables.

Train/Validation Split

A time-based validation strategy was used instead of a random split.

The data was divided into:

- **Training:** January 2025 - September 2025
- **Validation:** October 2025

I used a time-based split because the final predictions are for a later period. Random splitting would mix earlier and later observations and would not represent this situation as well.

The training set contained 43,147 observations and the validation set contained 4,853 observations.

Model Selection

CatBoost Regressor was selected as the main model.

CatBoost was suitable for this dataset because it can work directly with categorical variables and can handle missing numerical values. This was useful after treating invalid non-positive weight values as missing.

The final model used:

- 1,000 maximum iterations
- learning rate of 0.05
- depth of 8
- RMSE loss function
- early stopping based on the validation set

Validation Results

The model achieved the following results on the October validation set:

MAE - 119.11

RMSE - 647.93

R^2 - 0.8203

The MAE indicates that the model's average absolute prediction error was approximately $119.

The RMSE was substantially higher than the MAE. Further error analysis showed that this was mainly caused by a small number of extreme high-rate observations. These rare cases produced very large individual prediction errors and therefore had a much stronger effect on RMSE.

Error Analysis

Looking at the worst individual errors: the biggest misses were on unusually high “posted_rate” loads - several in the $14,000-$19,000 range that the model predicted at more like $3,500-$6,000.

These weren't concentrated in one equipment type. Dry Van and Flatbed had similar average error, Reefer was a bit higher. So this looks less like an equipment-category problem and more like a handful of unusual high-rate loads that the available features just don't fully explain.

Feature Importance

The CatBoost feature importance analysis showed that distance was by far the most important feature:

Feature	Importance
distance	86.37%
equipment	4.29%
weight	1.65%
market_index	1.42%
month	1.39%
quote_signal	1.07%
pickup	0.99%
delivery	0.91%
pickup_lat	0.69%
route	0.63%
pickup_lon	0.37%
day_of_week	0.23%

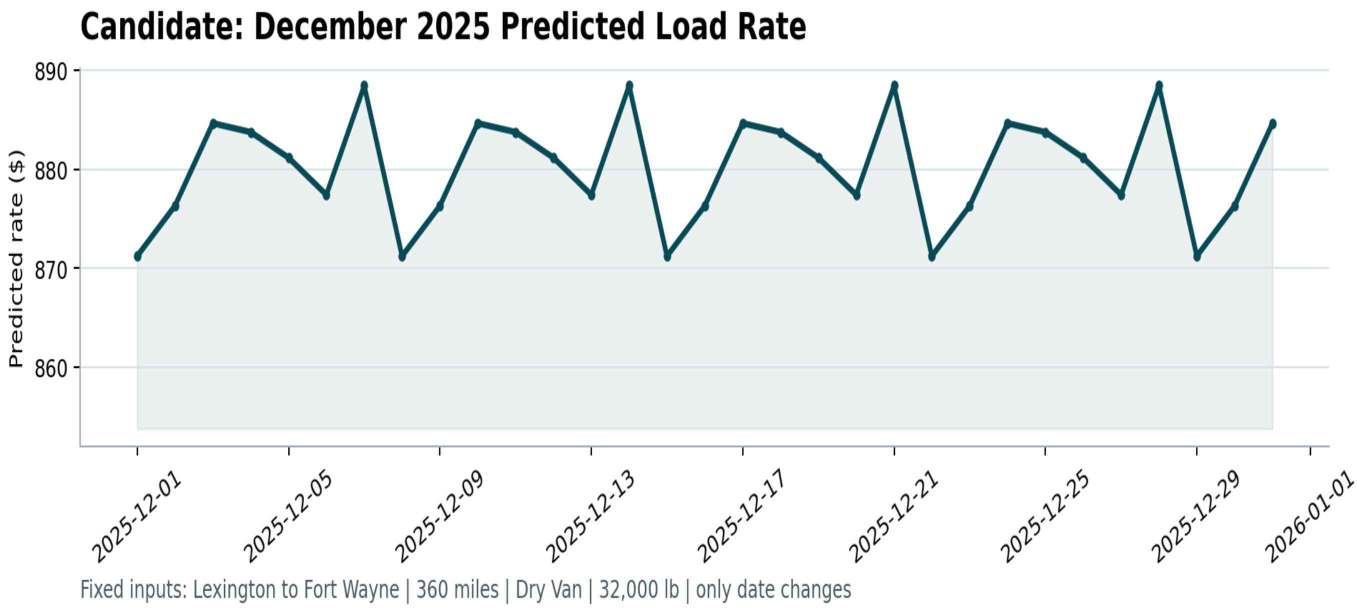
This lines up with the correlation analysis from earlier — distance was already the standout at 0.91.

Overall Conclusion

The CatBoost model gives a solid baseline for predicting load rates. I handled the data quality issues without stripping out observations that didn't need to go, and the time-based validation gives a realistic read on how the model would perform going forward rather than an optimistic one from random splitting.

The clearest weakness is the handful of very high-rate loads ($15k-$19k) where the model's predictions land well below the actual rate. I kept those records in rather than filtering them out — there wasn't a good reason to treat them as bad data, just hard cases.

Based on the validation numbers and the error analysis, I considered this model good enough to generate the final predictions for data/validation.csv.



Scorer results.